Pediatric Respiratory Distress, Failure or Arrest

1. Follow General Pre-hospital Care-Treatment Protocol.
2. Pediatric patients (< 14 years) utilize MI MEDIC cards for appropriate medication dosage. When unavailable utilize pediatric dosing listed within protocol
3. Assess the patient's airway
   A. If unable to ventilate patient after airway repositioning refer to Foreign Body Airway Obstruction-Treatment Protocol and/or Airway Management-Procedure Protocol
   B. Consider anaphylaxis refer to Allergic Reaction/Anaphylaxis-Treatment Protocol
4. Allow the patient a position of comfort that also maintains an open airway.
5. Titrate SpO2 to 94%
   A. Have a parent assist with oxygen via blow by or mask support.
6. Airway should be managed by least invasive method possible.
7. Suction secretions if needed.
8. Consider CPAP if appropriate size available, follow CPAP-Procedure Protocol
9. Do not delay transport for interventions.
10. Attempt vascular access only if necessary for patient treatment.

Suspected Bronchospasm (Wheezing):

1. Assist the patient in using their own albuterol Inhaler, if available and medication has not expired and is prescribed to patient.
2. Administer albuterol 2.5 mg/3ml NS nebulized (Per MCA selection may be EMT skill) per Medication Administration-Medication Protocol
3. Consider CPAP if appropriate size available, follow CPAP-Procedure Protocol
4. In cases of respiratory failure administer epinephrine auto-injector
   A. If child appears to weigh less than 10 kg (approximately 20 lbs.), contact medical control prior to epinephrine if possible.
   B. If child weighs between 10-30 kg (approximately 20-60 lbs.), administer pediatric epinephrine auto-injector IM.
   C. Child weighing greater than 30 kg (approximately 60 lbs.), administer epinephrine auto-injector IM.
5. In cases or respiratory failure administer epinephrine 1 mg/ml IM (per MCA selection may be BLS or MFR skill).

NOTE: BLS not carrying epinephrine auto-injector MUST participate in draw up epinephrine.

   A. If child appears to weigh less than 10 kg (approximately 20 lbs.), contact medical control prior to epinephrine if possible.
   B. If child weighs between 10-30 kg (approximately 60 lbs.), administer epinephrine (concentration of 1mg/1mL) 0.15 mg (0.15mL) IM
   C. Child weighing 30 kg or greater; administer epinephrine (concentration of 1mg/1mL) 0.3 mg (0.3 mL) IM

6. Per MCA selection, administer prednisone 50 mg PO to children > 6 years of age (if available per MCA selection).
   A. If prednisone is not available, patient is < 6 years of age, or patient is unable to receive medication PO, administer methylprednisolone IV/IO/IM:

i. Pediatrics: 2mg/kg

Appropriately trained EMT-Specialists (AEMT) may administer prednisone (PO) and methylprednisolone (IV/IO/IM) as specified in this protocol.

Stridor/Suspected Croup:

1. Croup is most common in children 6 months to 6 years of age
2. Commonly associated with recent upper airway infection or fever
3. If foreign body is suspected, and unable to be removed contact Medical Control prior to administration of nebulized racepinephrine/epinephrine See Foreign Body Airway Obstruction-Treatment Protocol
4. Consider humidified oxygen
5. If patient presents with stridor at rest without suspected airway obstruction administer nebulized epinephrine per MCA selection (Medical Control contact not required):
6. Do not delay transport.

Respiratory Failure or Arrest:

1. Ventilate the patient using an appropriately sized BVM with supplemental oxygen.
   A. Chest rise is the best indicator of successful ventilation.
   B. Ventilate at a rate appropriate for the patient:
      i. Infant: 30 breaths per minute
      ii. Child: 20 breaths per minute
   C. Utilize capnography per End Tidal Carbon Dioxide Monitoring-Procedure Protocol to maintain end tidal CO2 35-45 mm Hg.
2. Bag Valve Mask is the preferred method of ventilation for kids under 8 years old.
   A. When unable to ventilate with BVM and basic airway adjuncts, consider advanced airway see Airway Management-Procedure Protocol
3. If opioid overdose is suspected, administer naloxone according to MI-MEDIC cards. If MI-MEDIC is unavailable, administer naloxone per Opioid Overdose Treatment and Prevention-Treatment Protocol.
4. Monitor EKG and refer to Pediatric Crashing Patient/Impending Arrest-Treatment Protocol or appropriate cardiac protocol as required.

Nebulized albuterol administration per MCA selection

☑ EMT

MCA Approval of epinephrine auto-injector IM

☑ MFR

MCAs will be responsible for maintaining a roster of the agencies choosing to participate and will submit roster to MDHHS.

MCA Approval of draw up epinephrine.

☑ MFR

☑ BLS

Personnel must complete MCA approved training prior to participating in draw up epinephrine.
MCAs will be responsible for maintaining a roster of the agencies choosing to participate and will submit roster to MDHHS.

Additional Medication Option:

☑ Prednisone 50 mg tablet PO
(Children > 6 y/o)

MCA Selection

☑ Racepinephrine 2.25% inhalation solution via nebulizer
Administer by placing 0.5 mL of Racepinephrine 2.25% inhalation solution in nebulizer and dilute with 3 mL of normal saline.

☐ Epinephrine 5 mg (1mg/1ml) nebulized

Medication Protocols

Albuterol
Epinephrine
Methylprednisolone
Prednisone
Racepinephrine